Unit 3, Lesson 3: More with Powers of 10

Benchmark

MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.

Learning Target

- ☐ **I can use the Laws of Exponents to evaluate numerical expressions with a base of 10 and exponents less than 0.**

3.3.1 Warm-Up: Notice and Wonder

1. A picture of a bowling-ball pyramid is shown.

a. What do you wonder?

b. What do you notice?

c. Write a question about the pyramid that a 5th grader could answer.

3.3.2 Exploration: Powers of Ten

1. Use a scientific calculator to find the decimal value of each. Then, write the equivalent fractional value.

Power of 10	Decimal Value	Fractional Value
10^0		
10^{-1}		
10^{-2}		
10^{-3}		
10^{-4}		
10^{-5}		

2. What do you notice about the relationship between the power of 10 and its value?

3. Work with your partner to complete the table.

Power of 10	Fractional Value	Place Value of 1
10^0	1	Ones
10^{-1}		Tenths
10^{-2}		
10^{-3}		
10^{-4}		
10^{-5}		

4. **Work with your partner to describe how the exponent of the power of ten relates to place value.**

5. **Based on the pattern in the table, what is the value of 10^{-12}, written as a fraction?**

Negative Powers of **10**

A base 10 with a negative exponent can be rewritten as a fraction.

$$10^{-a} = \frac{1}{10^a}$$

The value of the fraction relates directly to the place value of the 1 when the quantity is expressed in its equivalent decimal form.

3.3.3 Try It: Make a Guess

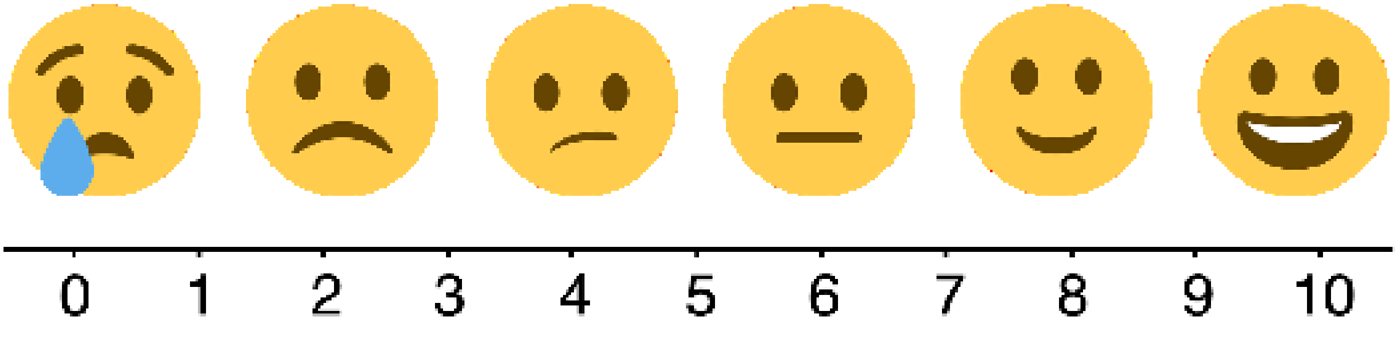
1. Without your partner or a calculator, make a guess on the value of each mathematical statement. Do not worry about being wrong. Remember, mistakes are a part of learning.

   Whenever possible, write the answer as a power of 10. Show your work or explain your thinking.

Mathematical Statement	My Guess	My Work or Explanation
$10^{-2} + 10^{-1}$		
$10^{-3} \times 10^{-8}$		
$10^{-1} - 10^{-3}$		
$\frac{1}{10^{-4}}$		
$(10^{-6})^2$		
$\frac{10^{-4}}{10^{-7}}$		

2. How confident are you in your guesses?

   Use the emoji scale to rate your confidence in your guesses.



3.3.4 Exploration: Using a Calculator to Check

1. Use a calculator to complete the following.

   a. Find the value of each mathematical statement. Then, if possible, rewrite the value as a power of 10.

Mathematical Statement	Value Found Using the Calculator	Value Written as a Power of 10
$10^{-2} + 10^{-1}$		
$10^{-3} \times 10^{-8}$		
$10^{-1} - 10^{-3}$		
$\frac{1}{10^{-4}}$		
$(10^{-6})^2$		
$\frac{10^{-4}}{10^{-7}}$		

   b. Compare your earlier guesses to the values you found for each expression, using a calculator.

   c. Discuss with your partner which mathematical statements could be rewritten using a power of 10 and which could not be. Summarize your findings.

2. Use your findings to name the law of exponents used to rewrite each mathematical statement.

Mathematical Statement	Value Written as a Power of 10	Law of Exponents
$10^{-5}(10^{8})$	10^{3}	
$\frac{10^{-11}}{10^{-4}}$	10^{-7}	
$\frac{1}{10^{-10}}$	10^{10}	
$(10^{-2})^{3}$	10^{-6}	
$10^{-5} \times 10^{-9}$	10^{-14}	
$\frac{10^{-3}}{10^{-7}}$	10^{4}	
$(10^{-6})^{-2}$	10^{12}	

3.3.5 Guided Instruction: Applying the Laws of Exponents

The Laws of Exponents can help when rewriting mathematical statements as a power of 10. Sometimes more than one law can be applied.

In the table from the Exploration, $\frac{10^{-11}}{10^{-4}}$ was written as 10^{-7}. Another way to write 10^{-7} is $\frac{1}{10^{7}}$. Each of these expressions represent the same value.

1. Write each mathematical statement in an equivalent form by applying one or more laws of exponents. Then, write the equivalent decimal value.

Mathematical Statement	Value Written as a Power of 10	Decimal Equivalent
$\left(\frac{1}{10^{-4}}\right)^{2}$		
$(10^{-11})^{2} \times 10^{27}$		
$\frac{10^{-12}}{10^{-5}} \cdot 10^{9}$		

Numbers like one-billionth, also known as a nano, can be written using powers of 10.

2. Write one-billionth, or $\frac{1}{1{,}000{,}000{,}000}$, as a power of 10.

3. One septillionth, 0.000000000000000000000001, is also known as yocto. Write one septillionth as a power of 10.

3.3.6 Wrap-Up: Reflection

1. Complete the table as you reflect on today's lesson.

<table>
<tr><td>

Today I deserve a

for …

</td><td></td></tr>
<tr><td>

Today I was feeling

about …

</td><td></td></tr>
</table>